

Quadcopter Ball-Transfer Drone - Final Engineering Report

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Quadcopter Ball-Transfer Drone
Course	Practice of Mechanical Engineering
Document Type	Final engineering report
Primary Role	Team lead, system designer, and integration coordinator
Original Source	drone-final-engineering-report.pdf (140 pages)

Executive Summary

This project developed a quadcopter capable of carrying and transferring a ball while maintaining sufficient flight stability for final evaluation. The mission required mechanical design, airframe layout, payload architecture, electronics integration, propulsion testing, and final-run risk control to work as one system.

My strongest contribution was keeping the team focused on system behavior instead of isolated subsystem success. Payload placement changed mass distribution, frame stiffness influenced vibration, and motor/propeller choices affected the available control margin.

The completed build earned full marks and a course grade of A+. The system also became a reference point for many peer teams, who looked to our architecture, testing approach, and integration decisions while developing their own builds.

Applicant Contribution Signals

- Led mission-level design decisions across airframe, payload, electronics, propulsion, and test strategy.
- Used thrust-to-throttle experiments to convert propulsion uncertainty into actionable design information.
- Separated signal-chain, flight-controller, structural, vibration, and motor issues during integration debugging.
- Prepared spare parts, rule checks, and fallback responses for a final evaluation with limited room for failure.

System Architecture

The drone was treated as a full robot system: airframe geometry, ball-transfer mechanism, power distribution, sensor/controller setup, and test procedure were evaluated together.

Design decisions were prioritized by their effect on stable flight and successful ball transfer, not by whether they looked elegant as individual components.

Testing and Validation

Propulsion behavior was studied through motor, propeller, ESC, battery, and throttle-response testing.

Flight tests were used to diagnose drift, vibration, crash risk, resonance, control setup, and mechanical fragility.

Final preparation focused on repeatability, spare assemblies, and controlled responses to likely failure modes.

Graduate Application Signal

This report shows the ability to lead a physical robotics project where mechanical design, embedded systems, flight behavior, and team decisions must be integrated under evaluation pressure.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Payload integration	Designed airframe and transfer mechanism as one system	Reduced late-stage instability after payload assembly	System architecture
Unknown propulsion margin	Ran thrust-to-throttle experiments	Turned motor behavior into design evidence	Experimental validation
Unstable flight symptoms	Separated signal, controller, vibration, and structure causes	Avoided treating every issue as tuning	Integration debugging
Final evaluation risk	Prepared spare parts and fallback plans	Made the final run less dependent on one perfect attempt	Engineering leadership

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.